GENG4411 Engineering Research Project Part 1
Project Proposal

Assessing climate injustice in the housing market

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April 2026

Signed Declaration

Project Summary

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Nomenclature

Chapter 1

Introduction, Literature Review and Project Objectives

1.1 Introduction

Following the onset of the pandemic, however, the Australian housing market began experiencing dramatic year-on-year growth, with Western Australia subsequently recording some of the most significant price increases nationwide (PropTrack, 2025). This rapid escalation has generated widespread concern that housing availability will not keep pace with demand. As a result, many first-home buyers are committing to increasingly unaffordable loans- according to KPMG modelling, these loans are now being serviced by 31% of household income, compared with 23% in 2019/20 (KPMG Australia, 2025). Yet, even with many people maxing out their borrowing capacity, the types of houses they can afford may still be less than desirable. Perth, a hot city, will continue to experience warm summers, Figure 1.1 compares the 5 most recent summers with all historical data. How Perth manages future heat stress is uncertain, and housing policy will be influential to an individual's heat exposure.

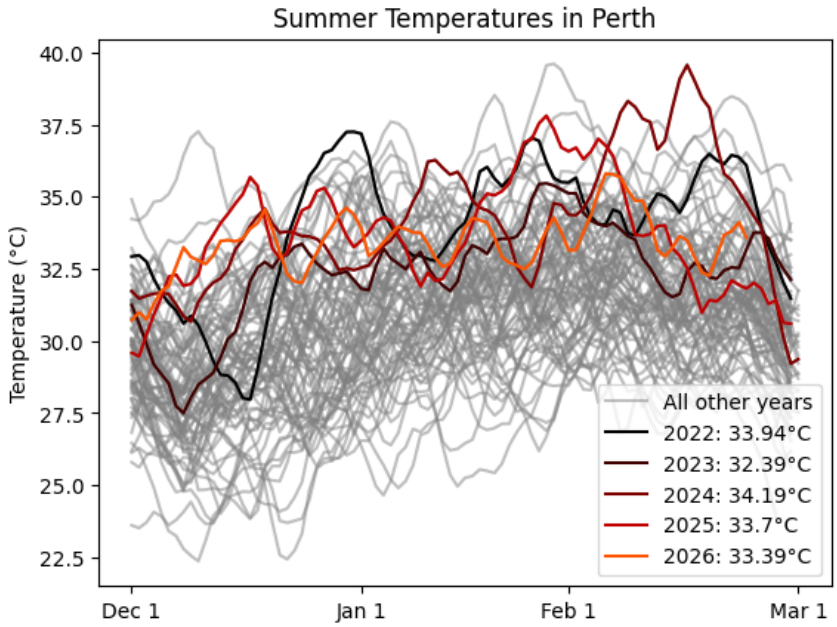

Figure 1.1: Rolling fortnightly averaged temperatures during Perth summer. Mean temperatures - including the hottest summer (2024) - given. Historical mean = 31.08°C

Data from Bureau of Meteorology, 2026

1.2 Literature Review (or Background)

A literature review will be conducted exploring four areas:

1. How built form responds to baseline and heatwave conditions, (determining whether to use a future climate model)
2. An appropriate methodology for determining air surface temperature and thermal comfort, (Justifying the use of TARGET)
3. Property market trends, (why is this an issue NOW?)
4. The human response to thermal stress, (why should we care?)

1.2.1 How built form responds to baseline and heatwave conditions

Built form - the concentration of buildings, paved surfaces, and other anthropogenic materials within the urban environment - plays a central role in shaping local thermal conditions. When present at sufficient density, these materials contribute to the urban heat island (UHI) effect, as urban areas exhibit systematically higher temperatures than their surrounding rural or less-developed landscapes. In the natural environment, evapotranspiration from vegetation and exposed soils offer environmental cooling. Additionally, the increased heat load from air-conditioning exhaust during these periods further contributes to UHI, however this will not be studied further in this project.

In normal conditions, this phenomenon occurs as harsh surfaces with a high volumetric heat capacity like concrete, pavement, and bitumen shift the energy balance such that more heat is absorbed throughout the day (Mohajerani et al., 2017). Overnight, this thermal load is released back into the environment, leading to warmer nocturnal temperatures such that the UHI-heatwave interaction is strongest overnight (Guo et al., 2026). During heatwaves, the geometries of urban areas further contribute to the entrapment of heat energy - the urban canopy decreases any atmospheric mixing, seen in Figure 1.2.

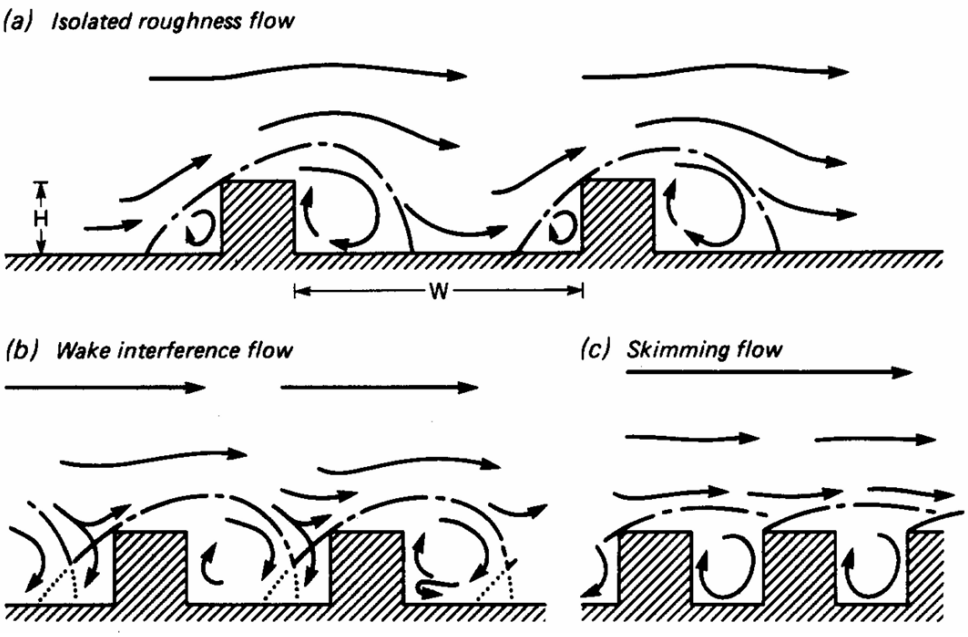

Figure 1.2: Flow regimes over urban canopy

(Oke, 1988)

As shown in Figure 1.1, temperatures have been observed to increase over the past several years. Going forward, Climate Council, 2024 predict an increase to days above 35° (totalling 26, up nine), an increase to days above 40° (totalling four, up two), and an increase to nights above 25° (totalling three, up two) by 2050 with current climate change action. Using future climate modelling as an input into the TARGET model uncovers the benefits of marginal increases to tree canopy cover, as previous studies have shown that the cooling influence of Green/Blue infrastructure is amplified/ ideal for heatwave forcings (Broadbent et al., 2018, Broadbent et al., 2019, Chen et al., 2024, Domeisen et al., 2022). Climate change is a global problem and for this reason, many feel hopelessness towards their own actions (Neale et al., 2023), however, with the increased sensitivity to marginal land cover changes discussed previously, individuals can improve their own environment.

TARGET can be run with future climate models, provided they can deliver a meteorological time series of air temperature, humidity, wind speed, air pressure, and incoming short and longwave radiation. Most future climate models do not provide the temporal resolution (hourly) required to run TARGET, however the study this project was modelled after, Nice et al., 2024 generated this information by retrofitting minimum, mean, maximum temperatures from the Coupled Model Intercomparison Project Phase 6 (CMIP6) projections (Eyring et al., 2016) to the ECMWF (European Centre for Medium-Range Weather Forecasts) ERA5 product, for 2030 and 2050 using emissions scenarios:

- 1.2-6: Limit warming below 2C,
- 3.7-0: Middle of the roadcurrent trajectory,
- 5.8-5: No policy (worst-case) scenario

Land surface cover data will be assumed to be in steady-state.

1.2.2 An appropriate methodology for determining air surface temperature and thermal comfort

There are many options for finding air temperatures in an urban environment, this section will step through each of their utility,

Satellite sensing

Satellites can only measure the surface temperature of surfaces they see (JOHN HAS SOURCE). Land surface temperature (LST) data from the most common satellite remote sensing project - Moderate Resolution Imaging Spectroradiomter (MODIS) - has a resolution of $500m/250m$ (Feng et al., 2015), compared to the block-scale ($20m \times 20m$) of the study. Further, in a study of the dynamics of LST and air temperature, it was found that the conditions of this study do not lend to LST being an appropriate proxy for air temperature (Naserikia et al., 2023).

Weather stations

The Perth metro region has a network of publicly funded weather stations from the Bureau of Meteorology (BoM) and Department of Primary Industry and Regional Development available (DPIRD, 2026), as well as a network of citizen weather stations (NETATMO, 2026). These stations offer insight into the climate of the greater Perth region, however their density is not sufficient to measure block-scale microclimates. Likewise, the locations of the commercial weather stations - typically in isolated areas, away from urban areas (Cantford, 1997) - does not allow comparisons between built form.

Sensors

Another option for observing urban microclimates is the deployment of dense networks of in-situ temperature sensors at targeted locations. For example, a London-based study deployed 39 sensors across parks and built-up areas to compare air temperature and thermal comfort conditions (Brousse et al., 2025). Each sensor cost approximately €335 and required calibration, installation, monitoring, and retrieval, a significant investment of both time and resources.

Beyond financial and labour constraints, sensor-based approaches are also subject to systematic location biases. Sensors must be mounted on existing infrastructure such as light poles, traffic signs, or buildings, and their placement is contingent on approval from the relevant authority; It is unlikely homeowners would allow sensors located inside their property. Scaling this approach to the scope of this study ($n = 50,000$), is therefore unfeasible.

Advanced models

(CHECK WITH JOHN)

TARGET

The Air-temperature Response to Green/blue Infrastructure Evaluation Tool (TARGET) offers reduced-complexity modelling of near-surface air temperature in the urban environment (Broadbent et al., 2019). It produces parcel-scale outputs (approximately 50–150 m × 50–150 m resolution) using site-specific land-cover data (REF NEARMAP). This allows simulations to extend across both public and private land, incorporating their respective surface characteristics into energy balance calculations.

Although TARGET was originally developed and validated for Melbourne (Broadbent et al., 2019), literature treats Melbourne comparably with Perth (Rogers et al., 2019). Additionally, it has been shown TARGET is suitable for other urban contexts (Chen et al., 2024), as it builds on the Local-Scale Urban Meteorological Parameterisation Scheme (LUMPS) (Grimmond & Oke, 2002) meaning the model captures key surface-atmosphere interactions without the risk of over-parameterisation. TARGET is therefore well suited to short-duration studies and permits the use of heatwave and future climate forcings, enabling consistent comparison across baseline and projected climate conditions.

1.2.3 Property market trends

Perth has experienced a steady population growth rate over the past several decades (Australian Bureau of Statistics, 2024-25) which has put increased stress on the housing sector (Crowe et al., 2025). Government responses have centred on rapidly expanding housing supply and increasing residential density around critical infrastructure, e.g., train stations; shopping precints, with current policy "Perth and Peel @ 3.5 million" (Western Australian Planning Commission, 2018) targeting 47% infill development and 168,000 new homes per year. One consequence of this shift has been a substantial increase in roof coverage relative to block size in new builds, as the market continues to reward house footprint over backyard space (Pandit et al., 2014) and consumer sentiment is largely against apartment-living (Leith van Onselen, 2024).

With limited land releases, the value of land has increased such that larger blocks are prohibitively expensive for many and they are often purchased by redevelopers to become sub-units. Given these options, home-buyers could be contractually committing to an environment with the built form characteristics which lead to high heat exposure. A United States study found the average person below the poverty line lived in areas with elevated UHI compared to those who earn more than two times the poverty line (Hsu et al., 2021). In the context of Perth,

the concern is these impacts are affecting a larger cohort of people - middle income households and young people - .

1.2.4 The human response to thermal stress

The idea that the environment fundamentally influences the human experience and human behaviours has been circulating since Ancient Greece, Rome, and imperial China - Hippocrates argued that factors like climate and topology have a profound impact on the physiology and health of native peoples; Confucius believed in strong linkages between Heaven, Earth, and Humanity. It should be no surprise that extreme hot temperatures and heatwave cause exacerbated rates of heat stress and higher rates of mortality. In fact it is the number-one cause of weather-related deaths globally; In June-August of 2003, 70,000 people died in a European heatwave event (World Health Organisation, 2024).

The extent to which other human outcomes are also influenced by increased heat-exposure is discussed in T. A. Carleton and Hsiang, 2016 (CAN I JUST COPY/PASTE THE GRAPHS?) which cites many studies that have found some link between climate variables and social and economic outcomes - increased temperature correlates to increased instances of aggression and conflict (Hsiang et al., 2011; Hsiang et al., 2013; Baylis, 2020; Ranson, 2014); decreased mathematics testing performance (NOT TRUE FOR ENGLISH/READING) (S. Graff Zivin et al., 2015), and decreased labour performance in high-risk industries (defined as agriculture, forestry, fishing, and hunting; mining; construction; manufacturing; and transportation and utilities industries.) (J. Graff Zivin & Neidell, 2014); GDP, productivity and income (Deryugina and Hsiang, 2014; Burke et al., 2015; Zhang, 2015). While these studies don't have Perth or Australian regions of interest and one could argue that a Perth population must have some higher level of tolerance to heat exposure due to climate adaptation, (T. Carleton et al., 2022) found that mortality rates during days over 35°, the mortality rate increases in Cold, Temperate, and Hot climates.

1.3 Project Objectives

Key discoveries that need to be made to prove/disprove the hypothesis

Where this project fall in to literature (Currently there is lots of talk about both climate resilience and housing affordability, there is limited papers linking them (some talk about extreme whether events). This paper will drive a strong link for "normal conditions")

Purchasing power scenarios

Presenting this data will in a user-friendly manner will be critically important in providing meaning to a wide audience.

Chapter 2

Project Process

2.1 Modelling Investigations

Modelling investigations often include model derivation and then implementation using a software package or written code. For modelling investigations this section might include:

- Model derivation (or modification)
- A description and diagram of the model configuration
- A description of any software tools / platforms to be used and/or created and reason(s) for choice
- Descriptions of the model input options to be considered (boundary conditions, constitutive behaviours, etc)
- A description of any model validation tests that will be done, i.e., how will you check that your model is producing or will produce reliable and accurate results?
- What are the models limitations?

Chapter 3

Timeline and Risk Management

Identify critical paths and dominoes in it.

Risks - installing/removing sensors, reliance on additional (discretional) funding...

Chapter 4

Progress to Date

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